

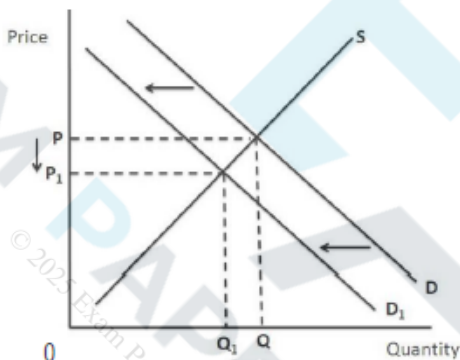
Mark Scheme

Q1.

Question Number	Answer	Mark
(a)	The only correct answer is C <i>A is not correct because the supply curve shift has been ignored</i> <i>B is not correct because this is the cost to the consumer</i> <i>D is not correct because this is the total amount received by the producer</i>	(1)

Question Number	Answer	Mark
(b)	Knowledge 1, Application 1, Analysis 1 Knowledge/Understanding 1 mark for identification of opportunity cost e.g. the next best alternative forgone. Application 1 mark for specific reference to government spending e.g. schools and hospitals. Analysis 1 mark for analysis of problem e.g. the economic and social case for a subsidy should be judged carefully on the grounds of efficiency and fairness. Government subsidies inevitably carry an opportunity cost and in the long run there might be better ways of saving energy.	(3)

Q2.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 2, Analysis 2 Knowledge 1 mark for definition of revenue or may identify clearly in diagram (1) Application 1 mark for using source data or linking ban to reduced consumption (1) and 1 mark for diagrammatic link to change in revenue (1), e.g. “can reduce tobacco consumption” (1) leading to decreased revenue as shown in diagram (1) Analysis Diagram showing demand shifts left or becomes less price inelastic (1) and new equilibrium (1)  (5)	

Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 2 - Application – price of e-cigarettes per nicotine cheaper - Data on PED suggests some rational response to price change e.g. 10% rise causes 4% fall in demand (Extract A) Reasons rational - Smokers computationally aware of financial gains of switching - Smokers informed of private health gains (Extract B Paragraph 2) - Reducing health external costs e.g. passive smoking (Extract A L15) - Smokers aware that regulations will increasingly make it difficult for them to gain utility of nicotine consumption through tobacco smoking NB Reasons for/against being rational may be presented as either KAA or EV (6)	

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 Reasons against being rational • Smokers lack information and computational skills to accurately assess full costs and benefits of switching. May underestimate long-term gains of switching (bounded rationality) • Information overload • Both products contain nicotine which is addictive and habit forming. • If e-cigarettes were not available they may quit smoking altogether • Switch may be only short-term (impulse buying) • E-cigarettes complements tobacco smoking resulting in an overall increase in nicotine consumption • PED for tobacco remains lower than e-cigarettes so a price rise in the latter may result in a switch back • Peer pressure (herd mentality) pushing smokers to make the switch	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
(c)	Knowledge 2, Application 2, Analysis 2 Knowledge/understanding 1 mark each for any two of the following: - definition or formula of cross elasticity of demand (1) - tobacco cigarettes and e-cigarettes have a positive cross elasticity of demand (1) - understanding of substitute relationship e.g. similar products (1) Application 1 mark each for any two of the following: - e-cigarettes seen as less harmful so become more attractive compared to tobacco cigarettes (1) - tougher regulations and growing health awareness on tobacco (1) - they are substitutes because they offer nicotine (1) - applying cigarettes to diagram showing impact on e-cigarettes (1+1) - relative price differences e.g. Figure 1 or tax rises on tobacco (1) - use of numerical example e.g. a 10% rise in the price of tobacco cigarettes results in a less than 10% rise in demand for e-cigarettes Analysis Up to 2 marks for any one of the following in linked development OR two limited examples of analysis (1+1): - the increase in the price of traditional cigarettes is resulting in the change in quantity demanded of e-cigarettes increasing - consideration of whether distant or close substitutes - diagram showing relationship between price of one of the product and demand for the other - magnitude/time	(6)

Question Number	Answer	Mark
(d)	Knowledge 2, Application 2 2 marks for each correct answer, e.g. - High-income countries = -4% / $+10\%$ (1) $= -0.4$ (2) or 0.4 (2) - Low-middle-income countries = -5% / $+10\%$ (1) $= -0.5$ (2) or 0.5 (2) NB Award 1 mark for a correct formula as an alternative for substituting correct figures on up to one occasion.	(4)

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 • Identification of determinants of price elasticity of demand • Application – PED tobacco cigarettes -0.35; PED e-cigarettes -1.9. Reasons why tobacco cigarettes being relatively price-inelastic, e-cigarettes relatively price-elastic e.g.: • habitual consumption • tobacco broadly defined • proportion of income spent on tobacco cigarettes lower than switching cost of buying reusable e-cigarette kit • number of close substitutes • tobacco addictive necessity, e-cigarettes luxury (not yet addicted to). NB to access Level 3 candidates need to consider tobacco and e cigarettes.	(9)

"Assess" was used as the command word for this question instead of "Discuss". In order to meet the expectations set out in the sample material the 10 mark level descriptors have been mapped on to the 15 mark levels as below on this one occasion. See Appendix 1 for full details.

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	7-9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • significance of determinant • e-cigarettes habit forming • tobacco cigarettes narrowly defined • proportion of income spent on tobacco cigarettes is rising • becoming closer substitutes • tobacco addiction waning as lost glamour and health warnings, e-cigarettes nicotine addiction. • longer term response to rising tobacco prices	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Indicative content	Mark
(f)	Knowledge 4, Application 4, Analysis 6 Functions of the price mechanism to allocate resources: • rationing • incentive • signalling Market diagrams for tobacco and e-cigarettes, making use of supply and demand analysis. * Consumers with knowledge of the potential harm of cigarette smoking may quit smoking. * Smokers will maximise their private utility * Consumers will decide what is in their best interests * Reduction in smoking will happen automatically, without government intervention, as e-cigarettes become popular.	

	Candidates may refer to Adam Smith Consumers free to pursue own self interest Profit incentive will mean firms will want their consumers to live long – so move out of tobacco and into e-cigarette market. Without government regulations e-cigarettes market will be easier for new firms to enter thus improving competition. Competition between e-cigarette manufacturers to maximise profits will drive health improvements and lower costs making them more price competitive. Candidates may refer to Friedrich Hayek Resource allocation from tobacco to e-cigarettes will be brought about by market forces, which will be far superior to any state planning system. Individuals do not have perfect information on cigarettes but have the best knowledge of their own situation. The price mechanism acts as a communication network, markets are self-correcting. Government lacks sufficient information about everything happening in the cigarettes market to be able to make informed decisions.	(14)
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	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(f) continued	Evaluation 6 Failings of the price mechanism • Information gaps • Asymmetric information • Externalities • Peer pressure ignores utility • Habit forming distorts utility Possible different disadvantages or approaches : * While consumers may maximise their private utility $MSC > MSB$, with explanation. e.g. passive smoking/health care costs * Without government intervention the oligopolistic/monopolistic power of cigarette firms, in the form of advertising and propaganda, will go unchecked Candidates may refer to Karl Marx Consumers exploited in past by misinformation on tobacco and due to being addictive they become dependent. Competition and the need to profit maximise will mean firms cut costs risking consumers health and increase prices thus reducing consumer surplus. Nature of free market is for powerful monopolies to emerge that can further exploit consumers.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Indicative content	Mark
(g)	Knowledge 4, Application 4, Analysis 6 Positive/negative effects of government intervention on tobacco cigarettes and or e-cigarettes with reference to market failure and using diagrams in various contexts: • Indirect taxation (ad valorem and specific) • Subsidies for tobacco replacements • Minimum cigarette price • Provision of information e.g. public health campaign • Regulation e.g. banning smoking in public places or advertising. • Direct provision – nicotine patches and e-cigarettes on NHS prescription Diagrams may be used to illustrate the above.	(14)

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Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
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Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(g) continued	Evaluation 6 Negative or unsuccessful impact of the intervention methods. Understanding of government failure as intervention that results in a net welfare loss – tobacco cigarette tax revenue exceeding tobacco related NHS expenditure. Causes of government failure: • Distortion of price signals – harm reduction instead of prevention, consuming e-cigarettes rather than quitting • Unintended consequences – e-cigarettes draw in new users acting as a gateway to smoking; regulating e-cigarettes protect tobacco from competition; tobacco smuggling • Excessive administrative costs – slow down e-cigarette innovation • Information gaps – little research on long term effects of e-cigarettes • Ineffective as PED < 1; diminishing returns of tax and regulations. Price inelastic demand diagram. Other evaluation approaches include: • Consideration of the extent of government intervention e.g. could it be more regulated • Free market approach might be more effective • Depends on the country being considered • Short run and long run	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Appendix 1

Part (e) directed candidates to "Assess" for 15 marks. In order to be in line with the support materials for this specification, this question should instead have directed candidates to "Discuss" for 15 marks. In order to ensure that all candidates received the fairest possible outcome, examiners were briefed on a levels-based marking structure for both command words. The body of the mark scheme includes the levels structure for the "Assess" command word as written in the paper. The levels structure that examiners considered for the "Discuss" command word is included below.

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 • Identification of determinants of price elasticity of demand • Application – PED tobacco cigarettes -0.35; PED e-cigarettes -1.9. Reasons why tobacco cigarettes being relatively price-inelastic, e-cigarettes relatively price-elastic e.g.: • habitual consumption • tobacco broadly defined • proportion of income spent on tobacco cigarettes lower than switching cost of buying reusable e-cigarette kit • number of close substitutes • tobacco addictive necessity, e-cigarettes luxury (not yet addicted to). NB to access Level 3 candidates need to consider tobacco and e cigarettes.	(9)

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Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • significance of determinant • e-cigarettes habit forming • tobacco cigarettes narrowly defined • proportion of income spent on tobacco cigarettes is rising • becoming closer substitutes • tobacco addiction waning as lost glamour and health warnings, e-cigarettes nicotine addiction. • longer term response to rising tobacco prices	(6)

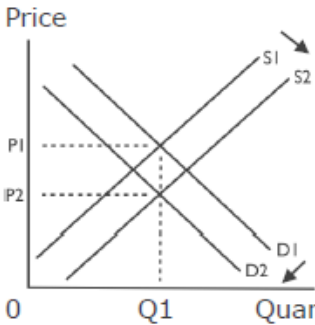
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Q3.

Question Number	Answer	Mark
(a)	The only correct answer is B A is not correct because difficulty in counterfeiting is an attribute of money C is not correct because divisibility is an attribute of money D is not correct because durability is an attribute of money	(1)

Question Number	Answer	Mark
(b)	Knowledge 1 Application 2 Award one function e.g. A measure of value (1) e.g. prices can be measured in satoshis (1) people can compare prices of goods and services (1) OR A store of value (1) when an individual saves bitcoin (1) they are confident they will receive the equivalent value back (1) OR A standard of deferred payment (1) when an individual loans another person bitcoin (1) they are confident they will receive the equivalent value back (1)	(3)

Q4.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 2, Analysis 2 Knowledge 1 mark for an increase in supply or a decrease in demand. This may be illustrated on a diagram Application 2 marks for using source data explaining change in supply and demand, e.g. • Estimated surplus (1) of approximately 7 million kg of uranium (1) • Kazakhstan's production has increased significantly/stockpiles (1) • Confidence in nuclear industry declined/Japan closed some of its nuclear plants (1) Analysis Accurate diagram – showing price fall (1) correct shift in demand and supply (1) e.g. 	(5)

Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 2 - Shows or demonstrates understanding of price elasticity of supply. - Reasons why uranium is price inelastic. - Diagram to show price inelastic supply. - Time. Two years to build a uranium mine in Spain, ten years in 'development', difficult to find in right concentrations. - Factor immobility e.g. uranium mine cannot be switched to other uses - Regulation. 'Have to clear extra hurdles' due to government concerns regarding negative externalities. NB KAA can be treated as Evaluation and vice versa	(6)

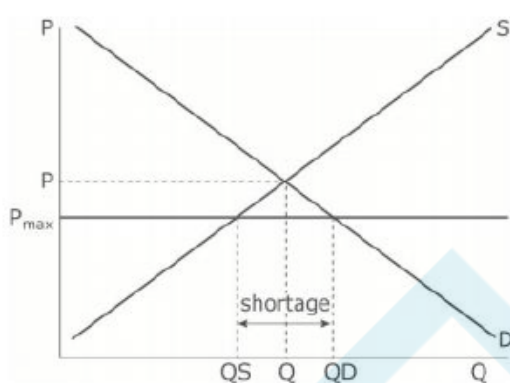
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 Reasons why uranium is price elastic: • Time. Ability to store stockpiles of uranium, Figure 1 surplus suggests and 'stockpiles until 2020'. • 'Uranium is a relatively common metal'. Extract A Line 2 • Distinction between short run and long run and its significance for elasticity of supply. By 2018 mine in Uranium producing '2.2 million kilos a year'.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
(c)	Knowledge 2, Application 2 Analysis 2 Award up to 3 marks for each reason: Knowledge/understanding • Identification of two separate reasons: inertia, computational problems, information gap, all providers increasing prices over time. (1) Application 1 mark for, e.g. • 'Irrational consumers pay the price'. (1) • Energy prices have risen by 158% in the last 15 years.(1) • Four million households are still on the most expensive energy rates. (1) Analysis Linked development of why not switching to a supplier offering lower prices (1) e.g. • Inertia (1) despite a decade of rising energy bills (1) as consumers do not have the time or motivation to switch (1).	(6)

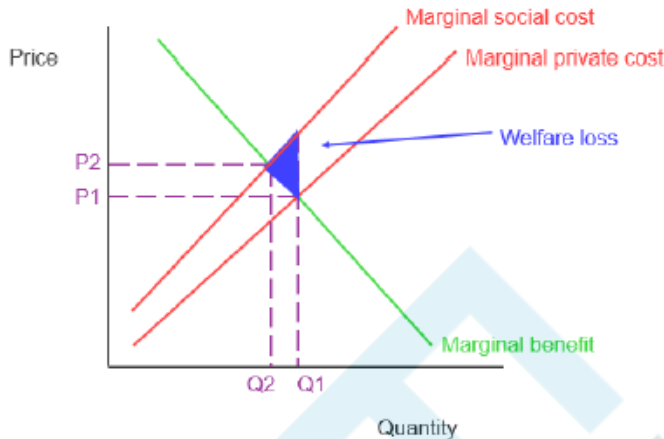
Question Number	Answer	Mark
(d)	Knowledge 2, Application 2 Knowledge/understanding 2 marks for e.g. Renewable energy is a resource made from something which can be renewed itself/infinite supply (1) Non-renewable energy is a resource which once exploited cannot be replaced/finite supply (1). Application 2 marks for, e.g. Energy generated by nuclear or coal is non-renewable (1) Solar or wind energy is renewable (1).	(4)

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 - Understanding of maximum price.  The effect of a maximum price on consumers: - Energy prices more affordable for consumers, end of 'decade of rising energy bills', extension in demand for energy or increase in quantity demanded (inefficient use). The effect of a maximum price on producers: - Contraction in supply or decrease in quantity supplied, power stations turned off 'undermine energy market', decline in producer surplus or revenue/profit. NB Level 3 responses must consider the effects on both energy consumers and producers. NB If no valid diagram then the maximum of Level 2 may be awarded	(9)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • It depends on whether and how far the maximum price is below the market equilibrium price. • It depends on the PES of energy. • It depends on the PED for energy. • Comment on the costs to consumers – loss of consumer surplus, energy shortages, 'higher prices in the future'. • Comment on the impact on producers – producer behaviour e.g. may move up to the price cap, reduce investment • Government failure e.g. distortion of price signals, unintended consequences	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Indicative content	Mark
(f)	Knowledge 4, Application 4, Analysis 6 External Cost:  Diagram(s) accurately drawn – with clearly labelled 'welfare loss' shaded triangle Explain costs to third parties in context. External costs: • Risk of nuclear accident – local schools and businesses close, impact on property prices. • High doses of nuclear radiation leaks increase the risk of cancer – loss of family income, employment, contribution to local economy, costs to NHS. • Cost of storage for future generations – opportunity cost of resources (land) tied to this, resulting in loss of income to farmers for example. • Opportunity cost of subsidies to nuclear industry, subsidy cuts to solar businesses. NB KAA can be treated as Evaluation and vice versa NB If there is no Diagram, max Level 3	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(f) continued	Evaluation 6 Reduced external costs in production of substitutes/external benefits may mean it is under-produced: • Curb greenhouse gas emissions which are causing climate change, by using nuclear, resulting in less flooding or storm damage reducing insurance costs to businesses. Extract C • Building new nuclear reactors allow positive local economic effects from job creation. • Provide 'affordable energy' helping businesses stay open thus protecting jobs and ending fuel poverty resulting in lower demands for health care. Other evaluation points might include: • Magnitude – depends on relative size of external cost. • Measurement difficult – putting a value on external costs. • Energy efficiency measures more preferable. • Short term/long term costs and benefits.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Indicative content	Mark
(g)	Knowledge 4, Application 4, Analysis 6 Reasons for government intervention e.g. to correct market failure Link intervention to reducing carbon emissions. Appropriate diagram(s) accurately drawn. Ways might include: • Indirect taxation (ad valorem and/or specific) on carbon emissions providers e.g. coal/gas power stations, or workplace parking, reference to taxes directed at getting people out of their cars • Subsidies to solar/wind/nuclear energy/cycling/public transport • Maximum price on low/zero carbon emissions providers and minimum prices on carbon emitters. • Tradable pollution permits e.g. EU emissions trading scheme. • Provision of information, energy efficiency measures. • Regulation e.g. shut down all coal-fired power stations by 2025, reduction in car emissions by using low emission zones. NB KAA can be treated as Evaluation and vice versa NB If there is no Diagram, max Level 3	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
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Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(g) continued	Evaluation 6 Why the ways given may or may not be successful e.g. • Distortion of price signals. • Unintended consequences, energy shortages. • Excessive administrative costs, pollution permits. • Information gaps, external costs of non-carbon based energy sources, questioning link between carbon emissions and climate change. • Government failure to provide, solar and wind power 'vary with weather', nuclear accidents. • Relative benefits/costs of government intervention versus market based solutions.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Q5.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 2, Analysis 2 Knowledge/understanding 1 mark for infinite wants and finite resources (1) Application 1 mark each for any one of the following for infinite wants: • Consumers of health care have a growing list of medical concerns for essential/non essential treatment/waiting list 4 million (1) • Ageing population (1) • New medical technologies(1) 1 mark each for any one of the following for finite resources: • Number of doctors/nurses (1) • Number of hospital beds available/bed occupancy rates 99% (1) use of Figure 2 e.g. • Waiting time has increased from 5.7% (1) to 10.7% (1) Analysis Up to 2 marks for any one of the following in linked development OR two limited examples of analysis (1+1): • Health resources are scarce (1) so Hospitals have to make choices about who to treat (1) • Decisions need to be made on what, how, why and for whom to treat (1+1) • Opportunity cost of making a medical choice/decision (1) • Government health priorities (1)	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2, Analysis 2 Knowledge/understanding 2 marks for e.g. • producers know more than consumers (1) or the information is incomplete or imperfect (1) • misallocation of resources (1) Application 1+1 mark for e.g. • risk of requiring health care is an unknown (1) • Doctors know more than patients (1) Analysis Up to 2 marks for any one of the following in linked development to market failure OR two limited examples of analysis (1+1): • Under consumption of healthcare (1) so patients' health conditions deteriorate (1) • Consumers overcharged for health treatment (1) as they are unaware of the true cost of health care (1) • Unnecessary treatment (1) as consumers unaware of health needs (1) • Patients lack of awareness of waiting times and NHS priorities (1)	(6)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 2 • Knowledge – public goods are non-excludable and non-rivalrous • Application - Extract C suggests public parks exhibit these features Reasons why they are public goods • 'free use' i.e. do not have to pay to enter • 'share the park with other users' • 'no exclusive use of the park' • Park run users 'share the park' with other users indicating one person using the park for a run does not have an impact on another person's enjoyment so non-rival. NB Reasons for/against being a public good may be presented as either KAA or EV	(6)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 Reasons against being public good - Local authorities planning on charging parkrun participants using public parks - Public parks already have exclusive areas e.g. booking tennis courts or football pitches - Parks may be rivalrous in consumption e.g. Hundreds of park run participants may get in the way of other users. - Parks are also excludable because they already have exclusive areas, gates can be closed at certain times	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
(d)	Knowledge 2, Application 2 £12 * 450 million (1) = £5.4 billion (1) £5.4 billion * 90% = £4.86 billion (1) £5.4 billion - £4.86 billion (1) = £0.54 billion £5.4 billion * 10% (1) = £540 million (1) NB Award 4 marks for a correct answer of £540 million/£0.54 billion. Award up to 3 marks for correct working but overall correct answer missing.	(4)

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 Demand shifting right and/or Supply shifting left e.g. Reasons for demand shifting right: • Ageing population, migration • Medical technological advances • NHS as a substitute less attractive, inferior good, waiting times. • Peer pressure Reasons for supply shifting left: • high management and administrative costs • salaries/wages rising due to doctor/nurse shortages • removal of subsidy/higher tax on private health care treatment • rising rents of private hospitals • rising cost of machinery in health care. NB For Level 3 KAA one relevant diagram.	(9)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	7-9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • Magnitude depends on how proportionately large the shift in demand or supply is as a key determinant • Price elasticity of demand may be more price elastic in the long run, given improvements in the NHS, discouraging further price increases • Price elasticity of supply may be relatively price inelastic, given health infrastructure, so price rise may be very significant • Other factors may reduce price e.g. government subsidies, productivity gains, healthier lifestyles.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-3	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	4-6	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Indicative content	Mark
(f)	Knowledge 4, Application 4, Analysis 6 Define excess demand. Application of excess demand to state funded health care. Government interventions: Factors influencing demand and supply (PES) that may result in the removal of excess demand (new hospitals built/doctors trained). Purpose of intervention with reference to market failure and using diagrams in various contexts: • government spending on direct provision increases supply • charging for NHS health care rations • indirect taxation (ad valorem or specific) on sugar/high fat content food/drinks or cutting taxes on healthier lifestyle options • subsidies on healthy food/gym memberships • maximum and minimum prices on healthy and unhealthy food/drinks Other methods of government intervention: • state provision of public goods e.g. public parks • provision of information e.g. 5 a day • regulation e.g. restricting sugar content • subsidising private treatment	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(f) continued	Evaluation 6 Government failure to reduce excess demand. Understanding of government failure as intervention that results in a net welfare loss; consumers face an under provision of food/drink that satisfies them i.e. marginal social benefits now exceed marginal social costs. Consumer choice restricted e.g. heavily subsidised healthy school meals and regulations against unhealthy options. Causes of government failure: • distortion of price signals • unintended consequences – charging for health care results in under consumption. • excessive administrative costs • information gaps – what is healthy, concerns about low sugar food/drink contents • time lag of the policies • magnitude depending on size of funding change. Irrational consumer behaviour - peer pressure.	(6)

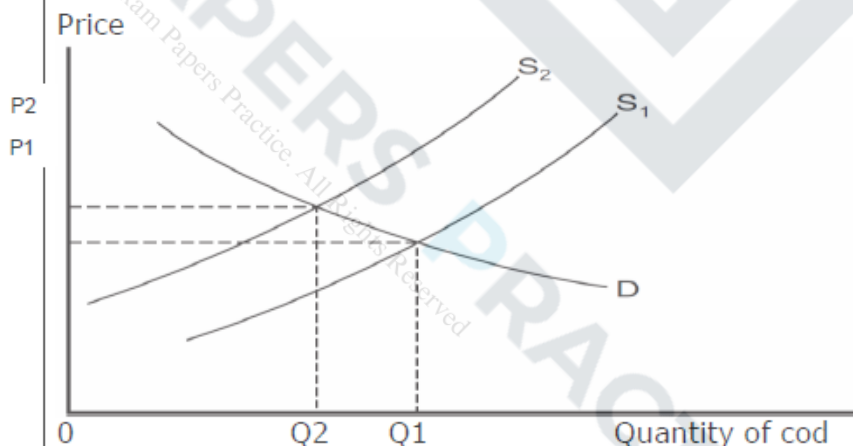
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Indicative content	Mark
(g)	Knowledge 4, Application 4, Analysis 6 Distinction between free market, mixed and command economies: reference to Adam Smith, Friedrich Hayek and Karl Marx Disadvantages of free market approach: • Free market under provision of health services, with large external benefits (SMB > SMC analysis), below social optimum. • Sick unable to work – loss of potential output, PPF. • Addressing concerns on affordability to access health care, insurance and information gaps e.g. preventative earlier treatments for elderly and children. • Misallocation of resources on administration of charges. • Profit incentive misallocating resources ahead of patient needs. • Asymmetric information on need to pay for treatment and quality of care. • Missing markets for elderly, children, remote rural areas.	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(g) continued	Evaluation 6 Functions of the price mechanism to allocate resources in private health sector or NHS: • rationing • incentive • signalling • 'Private health care increases choice for users, encourages competition and drives up standards of care.' • Mixed economy acts as a compromise between the advantages and disadvantages of a free market economy and a command economy. • Mixed economy features disadvantages of both a free market economy and a command economy and has no advantages of either. • NHS is more efficient and offers better care than private sector so why allow private sector to exist. • NHS waiting lists and lack of choice. • Information failures.	(6)
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Q6.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 2, Analysis 2 Knowledge 1 mark for original supply and demand diagram with original equilibrium Application 2 marks for using source data explaining change in supply e.g. • Cod catch limits cut (1) by 50% (1) • Reduced limited TACs agreed for Celtic Sea/Irish Sea/west of Scotland (1) • Relevant use of data from Figure 1 (1+1) e.g. Identifying the country that is overfishing (1) Identifying the amount of overfishing (1) Analysis Accurate diagram – showing price rise or candidate states price rise (1) correct shift in supply (1) e.g.  (5)	

Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 2 Reasons why fishing as much as possible is rational • This is completely rational from the perspective of a fishing company, which has revenues clearly linked to size of catch. • Shows an understanding of profit maximising behaviour. 'Private producers will act in their own self-interest and exploit common access resources to maximise profit.' • Figure 1 shows Sweden fishing at an excess of 52.4% in excess of TAC. • Using techniques to increase the size of the catch will help to minimise costs • 'Common access resources provide an opportunity for individuals to create profit, while spreading negative externalities to everyone else.' NB KAA can be treated as Evaluation and vice versa	(6)

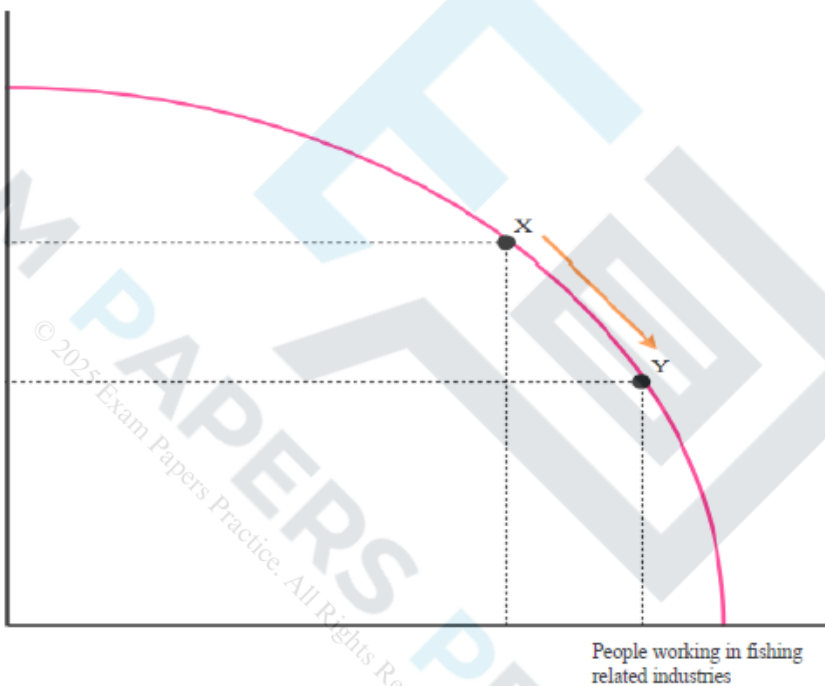
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 Reasons why fishing as much as possible is irrational: • ‘Exploitation is unsustainable and will lead to the depletion of resources; this is known as the tragedy of the commons.’ • Short run profits means that stocks are rapidly depleted so in the long run there are no resources for them to fish e.g. individual firms find that dwindling fish stocks mean their own revenue base is contracting. ‘90% of fish stocks are depleted.’ • It may be rational for firms to behave this way but also it is harmful to society of which those firms are a part e.g. negative externalities impact society as a whole	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
(c)	Knowledge 2, Application 2 Analysis 2 Knowledge and Analysis (1+1)x2 • Excludability (1) e.g. private ownership enables individuals to charge for the use of an asset (1) • Private ownership of resources (1) e.g. land, labour, capital and enterprise may be exploited by private individuals or firms (1) • Rivalry (1) e.g. private owners of resources can choose to buy and sell resources in order to generate profit (1) • Private ownership in a free market economy may lead to wage disparities or other inequalities (1+1) Application (1+1) 1 mark for use of Extract B e.g. ‘private producers will act in their own self-interest’ individuals ‘maximise profit’ 1 mark for use of own knowledge e.g. private ownership of a resource means someone is accountable for it (1) If someone owns something then they legally stop other people from using it (1)	(6)

Question Number	Answer	Mark
(d)	Knowledge 2, Application 2 1 mark for e.g. Identifies the correct formula (1) e.g. the formula for XED is $\% \Delta QD \text{ Good A} / \% \Delta P \text{ Good B}$ $-8 = x/2\%$ (1) $-8 \times 2\% = -16\%$ (1) (decrease in quantity demanded of bags of chips) $950\,000 \times 16\% = 152\,000$ (1) $950\,000 - 152\,000$ (1) = 798 000 If correct answer is given award all 4 marks. Award up to 3 marks for correct working but overall correct answer missing.	(4)

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 Impacts can be positive or negative. Understanding of specialisation. The number of workers in the fishing industry relative to other workers will be more in Scotland relative to the rest of the UK: People working in non-fishing related industries  People working in fishing related industries NB If there is no valid diagram - maximum level 2 KAA	

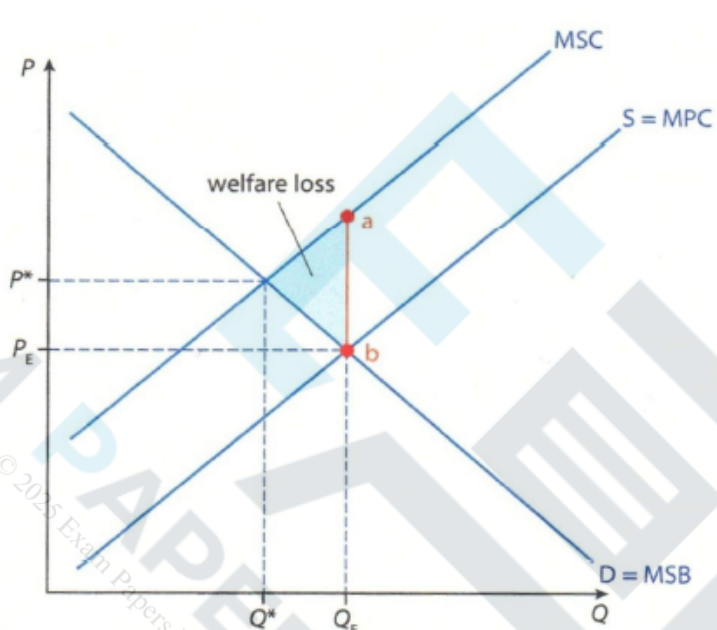
	Advantages might include: - Fishing fleets develop expertise based on years of experience - This will be the case for both labour and capital - Fishing ports have the infrastructure to process the catch and to distribute it to markets in the UK and overseas - Employment opportunities will be available and will grow as the industry develops - Specialisation enables training providers to specialise in developing skills for workers in the region - more resources can go in to	(9)
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	preparing for fishing industry without having to invest in training in other areas. • Spillover effects - skilled labour more freely available so if a new worker is needed they can benefit from training with a different company • Efficient use of resources – a region that specialises in fishing due to availability of resources e.g. good fishing grounds. • Creates jobs – people of Scotland will be able to find employment in the fishing industry • Scotland specialising in the production of fish can catch more and then trade with other countries who specialise in the production of other goods. • By catching more fish the economy can sell these, increasing income in the Scottish economy • By specialising and division of labour increasing output/ reducing costs of production the economy can improve income, living standards. NB KAA can be treated as Evaluation and vice versa	
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Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 There are many disadvantages of specialising in the production of goods and services to trade: • Over-exploitation of resources leaves economic agents over-exposed, leading to possible business failures and job losses • The reduction of the cod TAC could lead to major issues for Scottish fishing • External costs related to congestion and pollution • Demand for fish falls with the increased trend towards vegetarianism and veganism, the economy would be at risk • Workers could find it difficult to transfer to other industries if they are over-specialised • Current consumption may be unsustainable leading to reduced incomes for future generations	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Indicative content	Mark
(f)	Knowledge 4, Application 4, Analysis 6 Overfishing is a negative externality in production. Diagram e.g.  NB If there is no appropriate diagram - maximum level 3 KAA	
	Figure 1 shows the scale of the overfishing for different nations, e.g. Sweden's TAC is 52.4% above what the scientists advise. Multiple external costs are referred to in the Extracts. Extract B states that '90% of fish stocks are depleted.' The cod TAC should be reduced in order to reduce the extent of the overfishing. This would reduce the quantity caught from Q_E to Q^*. Possible external costs/third parties that might be affected - Fishing nets can damage wildlife, costing wildlife charities to rescue and rehabilitate animals - Fishing boats spill oil which has a negative impact	(14)

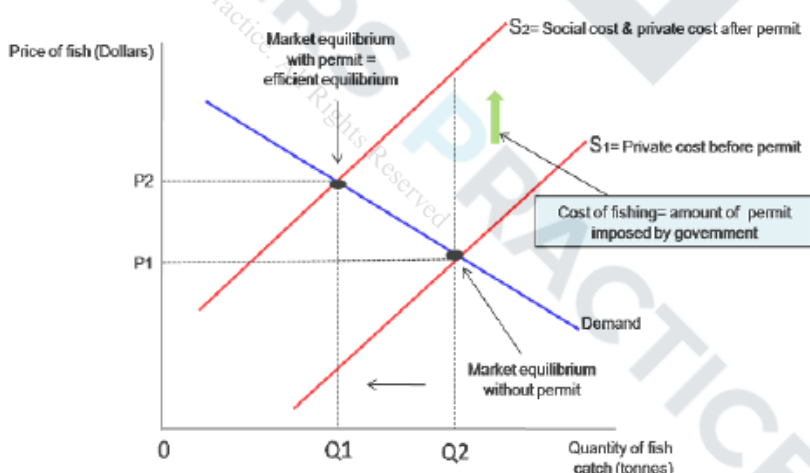
	on those swimming in the sea or wildlife, clear up for Environment Agency etc • Fishing industry throws any over catch back in to the sea, this can pollute the water • Smell from fishing industry can have a negative impact on those that live near the ports • Overfishing increases supply - lowers price and this has negative impact on manufacturers of other food items.	
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Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7-10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11-14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(f) continued	Evaluation 6 • Hard to put a value on external costs - how do you value harm to wildlife • Some impacts are slow to emerge • External benefits may exist - jobs in fish food processing plants, boat building and net making industries • Technology improves meaning risk of oil spills reduces • Net manufacturers are developing safer nets - fishing industry more aware of the need to dispose of nets appropriately • New method of fishing are reducing the external costs • TACs have helped to reduce the external costs	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5-6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Indicative content	Mark
(g)	Knowledge 4, Application 4, Analysis 6 Do not award policies involving changes in the total allowable catch (TAC) per country There are a range of policy options available to reduce over-fishing: Ways might include: • Provision of information, so that consumers can be encouraged to consume sustainably • Regulation e.g. designated areas for fishing allowing stocks to recover • Extension of property rights to international waters • Agreements on shares of fish that can be caught • Imposing indirect tax on fish to reduce consumption • Subsidy paid to firms to use safer technology • Minimum price for fish • Government permits awarded to fishing companies may be tradable • The answer may be supported using a relevant diagram e.g.	(14)

	 NB If there is no appropriate diagram - maximum level 3 KAA	
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Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(g) continued	Evaluation 6 Evaluation of policies might include; • Unintended consequences, such as illegal fishing • Excessive administrative costs • Information gaps, with consumers unaware of the extent of the damage being caused by over-fishing, not knowing level of tax to implement etc. • UK may increasingly over-fish since leaving the EU • Relative benefits/costs of government intervention versus market-based solutions • Government unaware of the right level of intervention	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Q7.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge and analysis (3): • Definition of opportunity cost as the next best alternative foregone (1) • Governments have to pay interest (1) at the expense of other expenditure (1) • Servicing means paying back interest and minimum capital requirements (1) e.g. to prevent a default, loss of confidence, other repercussions (1) • The debt is external (1), and in this case owed by governments as they have been borrowing from abroad (1) • Use of PPF to show opportunity cost may be awarded as analysis (1) Application (2 marks for 1 point, or 1 + 1): e.g. • Debt service payments above 20% of G (1) are unsustainable (Extract D) (1) • £20billion raised 2010-2015 (Extract D) (1) • Commodity prices soaring and foreign loans available' (Extract D) (1) • 'Heavy repayment schedules at a time of low commodity prices' (Extract D) (1) • Governments have taken to 'private debt too eagerly' (Extract D) (1) • 'Median level of debt risen from 34% to 48% in 2017' (Extract D) (1) is heavy because of 'low tax base' (Extract D) (1) • High levels of poverty – half of the 26 million people (1), so many feel the effects of reduced government expenditure (Extract E) (1) • 'non-existent tuna fleet' or other hidden debt (Extract D) or corruption (1) at the expense of other government spending (1) • Opportunity costs applied to any developing country, e.g. spending on poverty alleviation, education and health spending or recent cyclone April 2019 (1). Award a maximum of 1 application mark if there is no reference to Extract D	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge/implicit understanding of two reasons why manufacturing firms stay small, (1+1) e.g. • Size of market/lack of access to export markets/low export earnings due to lack of demand • Absence of economies of scale, or insufficient incentive or ability to grow (apart from access to financial) • Problems of human capital, e.g. lack of educated workforce • Lack of infrastructure making expansion into new areas difficult, or small markets • Actions of foreign firms to foster dependency • Government or international actions to make it hard for firms to grow • Lack of TNC involvement in Mozambique/volatility of currency/capital flight • Firms want to remain close to customers- meet needs with specific goods. • Firms might be manufacturing niche products with inelastic demand and high-income elasticity • Small firms more able to manufacture and sell through selling online and have small assets. • Perfect / monopolistic competition characteristics, e.g. low barriers to entry • Non-economic factors, e.g. natural disasters, political tensions, drought • Cost of finance, cost of servicing loans, financial economies of scale and scope. Do not allow access to finance.	
	Application (1+1 or 2): e.g. • Further depreciation of the metical (Extract E) • Lack of economic stability (Extract E line 26) • Slowdown in FDI (Extract E) • Depressed global demand (Extract E) • Low levels of employment in the sector (Extract E) may imply lack of skilled workers • High interest rates (Fig. 2 and Extract E line 14) • High import costs (Extract E line 15) • Needs urgently to improve its investment environment and confidence (Extract E line 17) • Large scale emigration (Extract E) • High numbers of people 'below the poverty line' (Extract E line 6) mean the market/incomes are small • 'Economic dependence on South Africa' (Extract E) may limit prices/market power of firms.	

	Analysis (1+1) - Development of each point, up to 1 mark for each valid factor Evaluation (2 marks for any relevant point, two points 1+1): - The development of the gas fields (Extract E) will 'transform the economy' - Pick up in coal and electricity (Extract E) which are probably larger firms and may have impact on manufacturing - may consider whether this is because of choice or lack of opportunity - Long term 'prospects are promising' (Extract E, line 20) e.g. sustainable agreement, gas fields production in 2020s or other initiatives such as trade blocs NB Cost of finance, financial economies of scale etc are allowed. It is only ACCESS to finance e.g. unavailability of microfinance, savings gap, loss of external lenders which is excluded. NB Access to finance <i>may</i> be credited as part of the evaluation but not as KAA	(8)
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Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 • Microfinance meaning and role • Plugs the savings gap / Harrod-Domar model • Enables firms to be set up – fixed or start-up costs • enables firms to invest in human capital and overcome primary product dependency • Use of Figure 3 to illustrate size and importance of microfinance operations • Use of Extract F – e.g. ‘100 000 borrowers’, many new business start-ups which could not have gained finance from any other source’, ‘unfulfilled demand for microfinance’ • How borrowers benefit – small firms can be set up or expanded, allowing firms to gain capital and materials to set up businesses, provide more employment, lead to improved living standards • Borrowers lack collateral so cannot borrow from banks in the normal way • The money is in small amounts £200 to £300, paid back regularly and over a short period of time, on which basis the firm forms a business plan for which the lender will approve or advise • Microfinance allows firms to form without the stigma of aid or the dependence of FDI • Use of cost/revenue analysis to support arguments (perfect/monopolistic competitive market) • Macro analysis (AD/AS) must link back to borrowers, e.g. increased standard of living, allows firms to expand/invest NB For Level 3, answers must link to borrowers NB The answer can take the form that there are more costs than benefits as part of the KAA, using the benefits as the evaluation	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • Use of Figure 2 to show high base rates and Figure 3 to show average 75% APR on microfinance • Large potential for expansion (Extract F) • May encounter lack of demand as more than half of the population live below the poverty line • Weak infrastructure does not allow borrowers to access markets • Alternatives to microfinance could be considered, e.g. aid, FDI, debt relief • Reasons why microfinance can be damaging to borrowers, e.g. pressure on women (98% of borrowers), over-indebtedness, poor guidance given to borrowers, macro-finance is needed for economies of scale, people borrowing multiple times, for uses which are social not business orientated. • Large numbers of people setting up similar businesses/highly competitive e.g. food stalls, so profits tend to be low/need for dynamic efficiency • Funds from private banks very different from NGOs – banks might be more cautious, and therefore more likely to succeed, might be better advice, support from specialists, softer terms from either institution might lead to moral hazard	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic influences may include: • Labour force and other demographic factors e.g. emigration of skilled/educated workers, disease, low life expectancy • Small firms with low levels of capital • Low levels of personal saving/high levels of personal debt • Primary product dependency • Volatility of commodity prices • Debt – although do not award access to credit and banking. Allow arguments based on dependency or opportunity cost, for example • Lack of infrastructure/other capital • Absence of property rights • Non-economic factors e.g. natural disasters, social problems • Joint ventures with global companies Macroeconomic influences may include: • Trade liberalisation – may refer to tariffs or trade agreements • Role of FDI e.g. the tuna fleet cover-up dries up FDI • Government policy failure, e.g. weak expansion in fiscal policy, high interest rates, lack of subsidies, privatisation, interventionist strategies, e.g. management of exchange rate, buffer stocks • Actions of other international institutions e.g. IMF or NGOs • Macroeconomic stability, e.g. rate of inflation (might refer to Figure 2), rate of economic growth, exchange rate (metical at a 'new low')	



	Possible evaluation points include: • Degree to which the factors are inevitable • Signs of improvements in passages • Some factors are temporary, e.g. fall in commodity prices such as coal and drought affecting production and exports to other countries within Africa NB Do not award KAA for arguments based on ACCESS TO credit and banking, although this approach is permitted as part of the evaluation NB for a Level 4 response there must be micro and macro factors (s). There must be reference to both growth and development for Level 4	
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	PPD/ Harrod-Domar/ debt/ aid/ debt relief/ cost of credit/ non-economic factors/currency issues / competitiveness index can be used as micro or macro.	(25)
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Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9–12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13–16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic factors. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8 Evaluation 9 Microeconomic effects may include: • On people with debt (better off) and savings (worse off) • On people with fixed incomes – may refer to falling real incomes • On firms' profits, e.g. that face rising costs and if they cannot raise prices they will face falling profits, or falling sales as consumers delay purchases. This might be illustrated with cost/revenue diagrams. • On firms' costs if they face rises in interest rates as a policy corollary of high inflation, which might lead to lower profits. This might be illustrated with cost/revenue diagrams. • Opportunity cost, e.g. effects on a government's spending if costs of servicing debts fall • Effect on labour markets e.g. people might work more hours to cover increased cost of living NB do not award fall in profits or revenue or rise in costs unless a valid chain of reasoning is given Macroeconomic effects may include: • International competitiveness and balance of payments, between African countries and with the rest of the world. Exports likely to fall and imports likely to rise • Distribution of income and wealth • Confidence and investment in financial markets • Policy responses, e.g. rises in interest rates and effects on AD • Shift in AS and/or AD may be used as part of the analysis • High inflation might deter Investment and FDI	

	NB do not award rise in unemployment/fall in growth unless a valid chain of reasoning is given Possible evaluation points include that the effect depends on: • Whether wages rise in line with inflation • The degree of inflation, and the policy and international responses. • The PED of countries buying the African exports • The PED of importing countries as imports become relatively cheap	
	• Whether African countries are trading with each other the effects of inflation may be cancelled out e.g. use of Figure 2 to show neighbouring countries with higher inflation rates • Whether asset prices rise as to the redistributive impact • Depends on whether this is a long-term inflation problem- could be short term e.g. linked to commodity price rises. Only a problem if it is sustained. • The figures are in one month in July 2017 and may not reflect inflation story for full year. Time series data may be more useful. • Depends on the type of inflation – e.g. cost push or demand pull. • Impact on exchange rates e.g. higher inflation may erode the value of the currency. This might cancel out the impact on competitiveness NB for a Level 4 answer there must be reference to the context of African countries (not just one) NB for a Level 4 response there must be micro and macro effect (s). Costs/ shift in PPF/ currency/labour markets/competitiveness etc. can be used as micro or macro.	(25)

Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
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Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q8.

Question Number	Answer	Mark
(a)	The only correct answer is C <i>A is not correct because Statement 1 is Positive, using testable data</i> <i>B is not correct because Statement 1 is Positive, using testable data</i> <i>D is not correct because 'ought' in Statement 2 is Normative</i>	(1)

Question Number	Answer	Mark
(b)	Application 2 Application $11\,000\,000 / 155 (1) = 70\,968$ or $11\,000\,000 / 156 (1) = 70\,512$ Award 2 marks for an answer within the range 70 500 and 71 000.	(2)

Question Number	Answer	Mark
(c)	Knowledge 1 Knowledge/Understanding 1 mark for definition, e.g. • centrally planned • government run • no or limited private sector • government allocates most/all resources.	(1)

Q9.

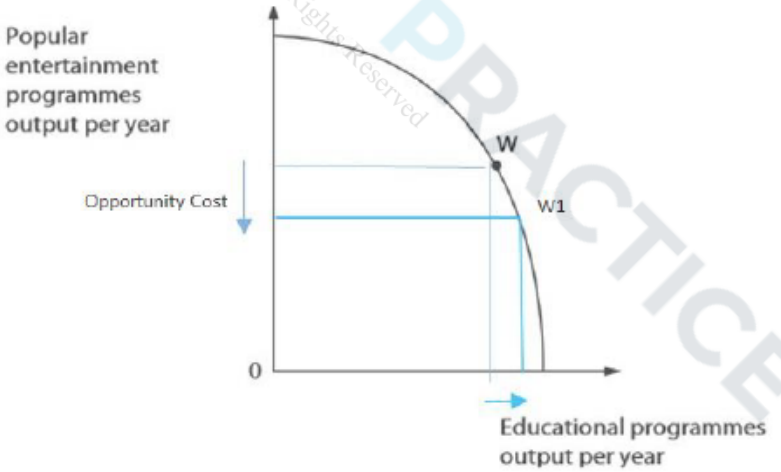
Question Number	Answer	Mark
(a)	Application 2 Application $PED = \% \Delta Q_d / \% \Delta P$ price rise = $5/60 \times 100 = 8.3\%$ (1) or $-5.4\%/+8.3\%$ (1) $PED = (-)0.65$ Award 2 marks for $(-)0.65$ NB if answer given is between $(-)0.7$ and $(-)0.6$ award 2 marks.	(2)

Question Number	Answer	Mark
(b)	The only correct answer is A <i>B is not correct because 'population' is a non price determinant of demand not PED determinant</i> <i>C is not correct because 'cost' is a non price determinant of supply not PED determinant</i> <i>D is not correct because 'expected rise in the price' is a non price determinant of demand not PED determinant</i>	(1)

Question Number	Answer	Mark
(c)	Knowledge 1 Knowledge/understanding 1 mark for definition: 'all other things being equal' (1) OR All other factors remain unchanged (1)	(1)

Q10.

Question Number	Answer	Mark
(a)	Knowledge 1 Knowledge/understanding 1 mark for definition, e.g. • Measure of the welfare (utility) that people gain from consuming goods and services • Difference between the total amount that consumers are willing and able to pay and the total amount that they actually pay. • Diagram showing area under the demand curve and above the market equilibrium price.	(1)

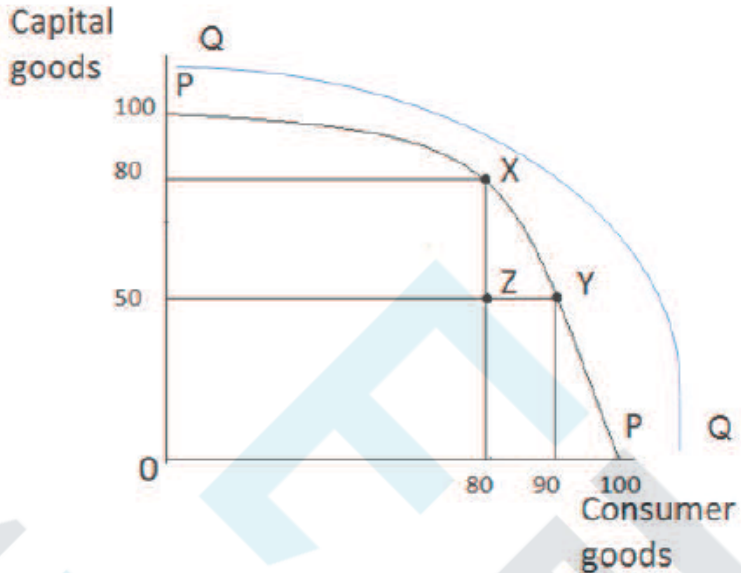
Question Number	Answer	Mark
(b)	Application 2 Application 2 marks for, e.g. An additional point to the right of W indicating movement of resources towards educational programmes (1) and the resulting opportunity cost, showing a fall in output of popular entertainment programmes (1). The following diagram earns 2 marks: 	(2)

Question Number	Answer	Mark
(c)	The only correct answer is C <i>A is not correct because this is providing an example of government finances rather than being non-excludable or non-rivalrous as a definition of a public good.</i> <i>B is not correct because this is confusing popularity with public goods, rather than being non-excludable or non-rivalrous as a definition of a public good.</i> <i>D is not correct because this is confusing benefits to third parties as members of the public rather than being non-excludable or non-rivalrous as a definition of a public good.</i>	(1)

Q11.

Question Number	Answer	Mark
(a)	Knowledge 2 Knowledge/understanding 1 mark for partial definition, e.g. • maximum potential output (1), OR • given current resources available (1). 2 marks for full and accurate definition, e.g. • a curve showing the maximum potential output (1) assuming all available resources are used fully (1).	(2)

Question Number	Answer	Mark
(b)	C	(1)

Question Number	Answer	Mark
(c)	Knowledge 1  New PPF drawn above line (1).	(1)